CineOrbit: Visibility-Aware Landmark Selection for Smartphone-Driven Camera Rotation Using Planned Pan Routes

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***Abstract*—A landmark chosen from the current camera frame may leave the field of view during a planned rotation, interrupting visual tracking. This paper presents CineOrbit, a low-cost smartphone-based platform in which JavaScript and OpenCV.js extract ORB features and an ESP32 drives a geared DC motor that rotates the camera. We propose Route-Conditioned Predictive Landmark Selection (RCPLS), which scores candidate landmarks by matching reliability, tracking persistence, movement cost, and predicted visibility over a predefined sequence of pan rotations. In a 1,000-route Monte-Carlo simulation of the geometric model (60° field of view, 2° per-step orientation drift), RCPLS reduced landmark-loss events from 1.54 to 0.79 per route (−49%) and raised in-FOV continuity from 94.4% to 95.4% relative to a reactive baseline, with 95.5% visibility-prediction accuracy that degraded to 91.1% at 6° drift. Matched-frame continuity did not improve (78.9% vs. 79.4%) because RCPLS sometimes prefers more visible but less distinctive landmarks. These are simulation results; hardware validation with the physical platform is specified but pending.**

***Index Terms*—visual landmark tracking, ORB features, predictive landmark selection, field of view, visual servoing, OpenCV.js, ESP32.**

I. Introduction

Camera-based perception underpins object detection, localization, visual tracking, inspection, and robot navigation. A common strategy is to identify distinctive visual landmarks and use them as reference features across successive images.

A key difficulty arises when the camera changes orientation. A landmark that is initially visible may drift toward the image edge or leave the field of view (FOV) during a planned rotation, and viewpoint change, illumination, blur, and partial occlusion further reduce the reliability of local features. A reactive approach that selects a landmark only from the current image does not account for the camera orientations expected later in a predefined movement sequence.

CineOrbit investigates this problem with a smartphone mounted on a motorized rotating platform. The smartphone performs image acquisition and processing: JavaScript and OpenCV.js extract ORB keypoints and descriptors, and a decision module generates directional commands that are sent over local Wi-Fi to an ESP32 microcontroller, which drives a geared DC motor through a motor driver.

The research component is the proposed Route-Conditioned Predictive Landmark Selection (RCPLS) method, which evaluates candidate landmarks against the planned camera-rotation sequence and combines predicted visibility with current tracking quality and persistence. The contributions of this paper are:

- a low-cost smartphone-and-ESP32 architecture for closed-loop visual camera rotation;
- a route-conditioned suitability score combining tracking reliability, predicted visibility, persistence, and movement cost;
- a repeatable experimental protocol for hardware evaluation; and
- a Monte-Carlo simulation study that quantifies the benefit and the limits of route-conditioned selection against a reactive baseline.

A. Problem Statement

Given landmark candidates observed by a rotating smartphone camera and a predefined sequence of camera rotations, select a candidate that is reliably trackable in current observations and likely to remain visible during upcoming camera movements.

B. Scope and Assumptions

The platform is stationary and only the smartphone camera rotates; the robot does not translate through the environment. The route is therefore a sequence of planned pan angles. Qualitative labels such as left, straight, and right must be mapped to calibrated angular displacements for geometric visibility prediction. The scene is assumed static. RCPLS is a proposed method and is not presented as a validated result.

II. Literature Review

This section reviews five strands of work that CineOrbit draws on—landmark selection, visibility-aware planning, visual servoing, local features, and active perception—and then positions RCPLS against them (Table I).

A. Landmark Selection for Planned Paths

Deng, Milios, and Mirzaian formulated landmark selection for path execution: given a planned path, choose which landmarks to detect and track along each segment so that sensing cost is minimized [1]. Their key idea, that

landmark usefulness depends on the route and not only on the current view, is the starting point of RCPLS. Their setting concerns a translating robot with a geometric map, whereas we consider a rotating camera and feature-based candidates.

B. Distinctiveness and Repeatability

Shimoda and Yamazaki rank visual landmarks by distinctiveness and repeatability [2]. A landmark must be distinguishable from its surroundings and re-detectable across observations. Related ideas appear in the "good features to track" criterion of Shi and Tomasi [9] and in appearance-based place recognition [12], where repeated texture causes ambiguous matches. RCPLS inherits this view through its reliability and persistence terms.

C. Visibility-Aware Planning

Lopez-Padilla and Murrieta-Cid add visibility restrictions to sampling-based path planning: the landmark must stay in the sensor FOV, be unoccluded, remain near the image centre, and change view orientation slowly [3]. They show that visibility can be handled as a planning constraint rather than an afterthought. Their method plans the robot's motion around a chosen landmark; RCPLS inverts this by taking the motion as given and choosing the landmark.

D. Visual Servoing and Field-of-View Constraints

Visual servoing closes the control loop on image measurements; position-based and image-based schemes and their trade-offs are surveyed in [10] and [11]. A well-known weakness of image-based servoing is loss of features at the image boundary. Kim and Oh combine position- and image-based servoing and use prediction to keep the target in view [4], and Albekairi *et al.* plan image-plane trajectories with FOV and collision constraints [5]. In both, the constraint acts on camera motion; none of these works selects among competing candidates using a known future camera route.

E. Local Features and Tracking

SIFT [13] set the standard for invariant local features, while FAST [14] and BRIEF [15] enabled fast detection and binary description. ORB [6] fuses them with orientation compensation and is the front end of ORB-SLAM [16], which shows that ORB is reliable enough for real-time mapping. MonoSLAM [17] similarly maintains a small set of landmarks over time. Comparison of binary descriptors uses Hamming distance, outliers are removed with RANSAC [18], and OpenCV [7] provides these components, including a JavaScript build suitable for a smartphone browser. Surveys of object tracking [19] note that tracking failures are dominated by occlusion, illumination change, and targets leaving the view, the last of which RCPLS targets.

F. Active Perception and Gaze Control

Frintrop and Jensfelt select attentional landmarks and control gaze actively to support visual SLAM [20], which is the closest conceptual neighbour to CineOrbit: a camera is moved so that useful landmarks stay observable. Their selection relies on saliency and current observations, not on a predefined camera route.

TABLE I. Comparison of Related Work and RCPLS

Work	Contribution	Difference from RCPLS
[1]	Landmark selection along planned paths	Map-based, translating robot
[2]	Distinctiveness and repeatability	No future-view prediction
[3]	Visibility-constrained planning	Plans motion; landmark fixed
[4], [5]	FOV-aware visual servoing	Constrains motion, not selection
[16]	ORB-based SLAM	Maps landmarks; no route-aware choice
[20]	Attentional landmarks, gaze control	Saliency, not planned route
RCPLS	Route-conditioned selection	Planned pan route sets the score

G. Research Gap

Existing work selects landmarks for a path [1], ranks them by repeatability [2], plans motion around a fixed landmark [3]–[5], or controls gaze using current saliency [20]. We found no work in the reviewed set that scores candidate ORB landmarks against a known future camera-rotation sequence on a low-cost smartphone platform. This is an observation about a selective survey, not a claim of novelty, which would require a systematic search.

III. System Architecture

CineOrbit comprises a smartphone camera and processing module, a decision and route-evaluation module, a local Wi-Fi link, an ESP32, a motor driver, and a geared DC motor on the rotating platform (Fig. 1).

Camera frame → OpenCV.js (grayscale, ORB) → Matching + candidate grouping → RCPLS scoring and switching → Command → Wi-Fi (HTTP) → ESP32 → Motor driver → DC motor → Camera rotation → (new frame)

Fig. 1. CineOrbit closed-loop pipeline. [Placeholder: replace with block diagram and prototype photograph.]

Acquisition and ORB extraction: the rear camera supplies frames to a JavaScript application; OpenCV.js converts them to grayscale and detects ORB keypoints and descriptors, which are matched across frames to find persistent features.

Decision module: candidate feature groups are scored using matching reliability, route-predicted visibility, persistence, and movement cost; a candidate is selected or retained under a switching rule (Section IV).

Communication and actuation: movement commands are sent to the ESP32 over local Wi-Fi, for example as HTTP requests, subject to network and browser security configuration. The ESP32 [8] drives the motor driver and geared DC motor. After each movement the smartphone captures a new frame and the loop repeats.

IV. Proposed Method: RCPLS

A. Predefined Camera Route

The route is a sequence of relative rotations $R = \{\Delta\theta_1, \ldots, \Delta\theta_N\}$, and the predicted orientation at step k is

$$\theta_k = \theta_0 + \Sigma_{j=1}^{k} \Delta\theta_j \quad (1)$$

The route must contain calibrated angular displacements. Orientation should be updated from encoder feedback or a calibrated speed-and-duration estimate; open-loop estimates may drift with load, voltage, and mechanical variation.

B. Feature Matching and Candidate Grouping

For frame I_t, ORB yields keypoints K_t and binary descriptors D_t. Descriptors of consecutive frames are matched by Hamming distance, ambiguous matches are removed with a nearest-neighbour ratio test, and geometrically inconsistent matches are rejected (e.g., RANSAC where a model can be estimated). Spatially consistent matches form candidate landmark regions; a candidate is a feature group, not necessarily a semantic object.

C. Tracking Reliability

For candidate L_i, let M_i be a normalized matching score and G_i the geometric consistency ratio:

$$G_i = N_{inlier,i} / N_{matched,i} \quad (2)$$

$$R_i = \alpha M_i + \beta G_i \quad (3)$$

with $\alpha, \beta \geq 0$. M_i requires a documented normalization range, and coefficients should be chosen by calibration.

D. Bearing and Predicted Visibility

Let u_i be the horizontal image coordinate of the candidate centre, c_x the principal point, and f_x the focal length in pixels. The bearing relative to the current orientation is

$$\varphi_i = \operatorname{atan}((u_i - c_x) / f_x) \quad (4)$$

giving a world-relative bearing $\psi_i = \theta_0 + \varphi_i$ and, at predicted orientation θ_k, a relative bearing $\varphi_{i,k} = \operatorname{wrap}(\psi_i - \theta_k)$. The candidate is predicted visible at step k when $|\varphi_{i,k}| \leq FOV_h/2 - \delta$, where δ is a safety margin. The route-weighted visibility is

$$V_i = \Sigma_k q_k \cdot 1(|\varphi_{i,k}| \leq FOV_h/2 - \delta) / \Sigma_k q_k \quad (5)$$

where q_k weights future steps and $1(\cdot)$ is the indicator function. The model assumes a static scene and approximately constant bearing; it ignores depth-dependent parallax, occlusion, and scene motion.

E. Persistence and Movement Cost

Persistence over a window of W frames is $P_i = T_i / W$, where T_i is the number of frames in which the candidate is tracked. A current-view movement cost penalizes candidates near the FOV boundary:

$$C_i = |\varphi_i| / (FOV_h/2) \quad (6)$$

In implementation this term should be bounded and extended with mechanical pan limits where applicable.

F. Suitability Score and Selection

$$S_i = w_R R_i + w_V V_i + w_P P_i - w_C C_i \quad (7)$$

The weights are non-negative. The selected landmark is $L^* = \operatorname{argmax} S_i$ over L_{valid}, the candidates meeting minimum matching, persistence, and visibility requirements. Weights, thresholds, and normalization are to be fixed by calibration and sensitivity analysis.

G. Landmark Switching and Recovery

To avoid oscillation between similarly scored candidates, a new candidate L_n replaces the current L_c only if $S_n > S_c + \tau$, where τ is a switching margin; a minimum-persistence condition may also apply. If no candidate is valid, the system enters a defined recovery state rather than issuing unverified tracking commands.

H. Motor Command Generation

The module computes the error between the selected landmark's image position and a reference position and maps it to left, right, or stop commands sent to the ESP32. Motor control should include motion limits, a timeout or watchdog, and an explicit stop behaviour. GPIO assignments and PWM configuration are documented only after final wiring and ESP32 Arduino Core version are confirmed.

I. Algorithm Outline

1. Initialize camera, OpenCV.js, ORB detector, route, and calibration parameters.
2. Capture a frame; detect ORB keypoints and descriptors.
3. Match against the previous frame; reject ambiguous and inconsistent matches.
4. Group consistent features into candidate landmarks.
5. Compute R_i, φ_i, P_i, V_i, and C_i per candidate.
6. Compute S_i; discard candidates failing validity conditions.
7. Retain the current landmark unless a candidate exceeds it by τ.
8. Generate and transmit a movement command to the ESP32.
9. Rotate, capture the next frame, and repeat until the route ends.

V. Experimental Design

This section defines the hardware evaluation protocol. No hardware measurements are reported in this paper; the simulation in Section VI evaluates the geometric model only.

A. Configurations

Configuration A (reactive baseline) selects candidates from current-frame ORB matching and tracking quality without route information. Configuration B (RCPLS) uses the same pipeline and adds predicted visibility over the rotation sequence. Both use the same camera, resolution, route, motor settings, environment, and matching thresholds.

B. Test Scenarios

- Stationary observation to establish baseline detection and matching behaviour.
- Small controlled rotations (gradual viewpoint change).

- Large rotations moving landmarks toward and beyond the FOV boundary.
- Predefined multi-direction routes comparing reactive and route-conditioned selection.
- Partial obstruction to examine switching and recovery.
- Varied illumination to examine ORB sensitivity.

C. Metrics

TABLE II. Hardware Evaluation Metrics

Metric	Definition
Tracking continuity	Tracked / evaluated frames × 100
Landmark loss frequency	Loss events per route
Matching reliability	Valid matches, inlier ratio per frame
Visibility prediction success	Correct / all predictions × 100
Processing time	Per-frame processing and selection time
Alignment error	Landmark vs. reference position
Motor response time	Command to observed motion

D. Data Collection Procedure

1. Mount the smartphone securely and fix the test environment.
2. Calibrate horizontal FOV, image centre, and pan-angle response.
3. Define and record a route with explicit angular displacements.
4. Run Configuration A; log detections, matches, landmark identity, losses, and commands.
5. Repeat the route under identical conditions with Configuration B.
6. Repeat trials; retain raw logs, timestamps, route definitions, and frames.
7. Compute each metric and compare configurations.

VI. Simulation Study and Results

Because the physical platform has not yet been characterized, we evaluated the RCPLS geometric model by Monte-Carlo simulation. These results test the model under stated assumptions; they are not measurements of the prototype.

A. Setup

Each of 1,000 trials draws a static scene of 12 candidate landmarks with world bearings uniform in [−110°, 110°] and latent tracking quality q ~ Beta(5, 2.5). The horizontal FOV is 60° with margin δ = 5°. The route has 8 pan steps of 15°–40° in random directions, clipped to ±100°, with 10 frames per step. The actual orientation equals the planned one plus a random-walk drift (σ = 2° per step); bearing measurement noise is 1°. A tracked landmark is matched in a frame with probability q + 0.1, reduced near the FOV edge. Both methods re-select from visible candidates after a loss. The baseline scores R, P, and C only; RCPLS adds V computed over the next three route steps (weights 0.35, 0.35, 0.15, 0.15 for R, V, P, C; τ = 0.05). All parameters are assumptions, not calibrated values.

B. Results

TABLE III. Simulation Results (1,000 Routes, Mean ± 95% CI)

Metric	Baseline	RCPLS	Unit
Loss events	1.54 ± 0.09	0.79 ± 0.07	per route
In-FOV continuity	94.4 ± 0.6	95.4 ± 0.7	%
Matched continuity	79.4 ± 0.8	78.9 ± 0.8	%
Visibility prediction	N/A	95.5 ± 0.2	%

Hardware metrics (processing time, alignment error, motor response) are not simulated.

RCPLS cut loss events by 49% (paired difference 0.76 ± 0.06 per route) and raised in-FOV continuity by 0.97 ± 0.10 percentage points. Matched-frame continuity did not improve; it was 0.5 points lower. In this model the matching probability is tied to latent quality, and RCPLS accepts somewhat lower-quality candidates that stay visible longer, so it trades per-frame match quality for fewer interruptions.

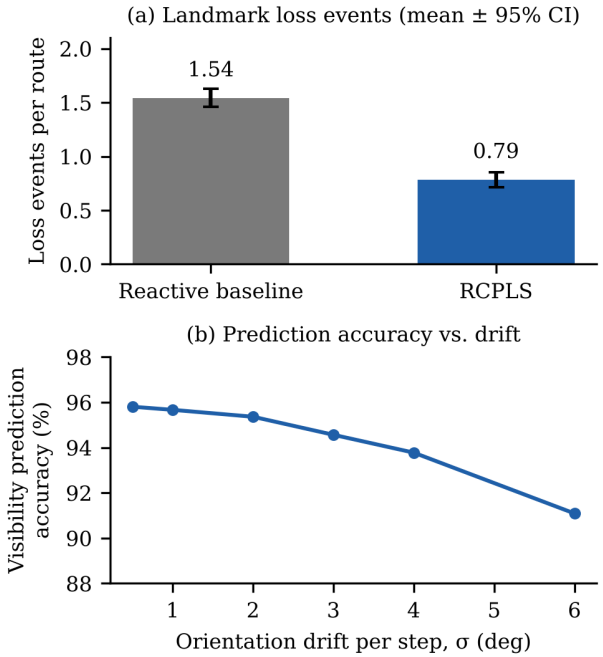


Fig. 2. Simulation results: (a) loss events; (b) prediction accuracy vs. drift.

C. Sensitivity to Orientation Drift

TABLE IV. Effect of Per-Step Drift σ (400 Routes Each)

σ (deg)	Pred. acc. (%)	Loss: baseline	Loss: RCPLS
0.5	95.8	1.65	0.83
1	95.7	1.46	0.75
2	95.4	1.46	0.78
3	94.6	1.61	0.81
4	93.8	1.63	0.97
6	91.1	1.61	1.04

Prediction accuracy falls from 95.8% to 91.1% as drift grows from 0.5° to 6°, and the loss-event advantage narrows from 0.82 to 0.57 per route. This supports the encoder-feedback recommendation of Section IV-A. A smaller margin δ raised prediction accuracy in a further

sweep (97.3% at $\delta = 0°$ vs. 91.6% at $\delta = 10°$), but that sweep scores accuracy against an unmargined FOV, so it should not be read as a recommendation to set $\delta = 0$. A weight ablation ($w_V = 0$, balanced, route-heavy) gave in-FOV continuity within overlapping 95% CIs (94.4%–95.1%), while loss events fell as the route weight rose (1.29, 0.87, and 0.75 per route), so in this setup the route term reduces interruptions without a clear continuity gain.

VII. Discussion and Limitations

The simulation indicates that future route information mainly helps by avoiding avoidable losses (−49%) rather than by raising matching quality, and that the gain depends on how accurately camera orientation is known. Predicted visibility must still be checked against observations on hardware: a high predicted score can fail because of occlusion, parallax, lighting change, or camera-angle error, so prediction quality should be reported separately from tracking continuity. The simulated gains rest on assumed distributions and a noise model chosen by the authors and may not transfer to real scenes. The stationary rotating setup isolates orientation and FOV effects but does not establish performance for a translating robot; richer settings may need depth estimation, visual odometry, occlusion reasoning, or encoder-based pose feedback.

Known limitations are: ORB features lack semantic recognition; the visibility model assumes a static scene and near-constant bearing; motor-based orientation estimates drift without an encoder; calibration errors affect bearing and FOV; occlusion, parallax, blur, illumination change, and repeated texture cause mismatches; browser camera access may need a secure context and cross-origin requests need server configuration; the simulation results are not hardware measurements, and RCPLS remains unvalidated on the prototype.

VIII. Conclusion and Future Work

We presented CineOrbit, a smartphone-based landmark detection and motorized camera-control platform built on OpenCV.js, local Wi-Fi, an ESP32, and a geared DC motor, and formulated RCPLS, a route-conditioned selection method combining ORB reliability, predicted FOV visibility, persistence, and movement cost. In a Monte-Carlo simulation, RCPLS halved landmark-loss events (1.54 to 0.79 per route) and slightly increased in-FOV continuity, but did not improve matched-frame continuity, and its prediction accuracy degraded with orientation drift. Confirming these effects requires the hardware experiments of Section V.

Future work includes running the hardware protocol, implementing the full RCPLS pipeline, calibrating camera and motor orientation, adding encoder feedback, improving grouping and geometric verification, handling occlusion, evaluating across routes and environments, and extending to dynamic route updates and mobile platforms.

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